Condensation and Metastability in the supercritical two-species zero-range process via resolvent and H^1-approximation

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Contents

1. Classical zero-range process
2. Two-species zero-range process
3. Main theorem

Classical zero-range process

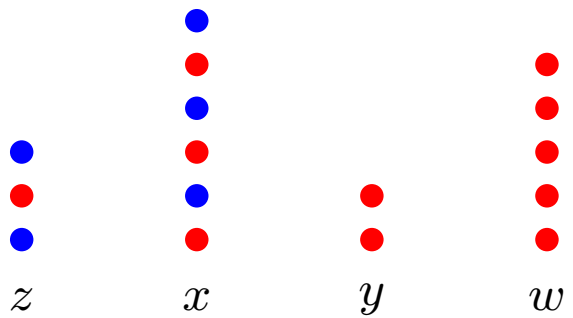
- **Random walk on finite state space S**
  - ⋆ Irreducible jump rate $r : S \times S \to [0, \infty)$
- **Configuration space** $\Omega_N = \{\eta \in \mathbb{N}_0^S : |\eta| = N\}$
  - ⋆ $\eta = (\eta(x))_{x \in S}$
  - ⋆ $\eta(x) \in \{0, 1, \ldots, N\}$
- **Dynamics** Particle jumps $x \to y$ at rate $g(\eta(x))r(x, y)$

$$g(n) = \begin{cases} \frac{\mathfrak{a}(n)}{\mathfrak{a}(n-1)} & \text{if } n \geq 1 \\ 0 & \text{if } n = 0 \end{cases}, \quad \mathfrak{a}(n) = \begin{cases} n^{\alpha} & \text{if } n \geq 1 \\ 1 & \text{if } n = 0 \end{cases}.$$

Two-species zero-range process

- **Two random walks on finite space** S
  - ⋆ Irreducible jump rates $r_1, r_2 : S \times S \to [0, \infty)$.
  - ⋆ Normalized stationary measure m_1, m_2, i.e, $\max_{x \in S} m_i(x) = 1$.
  - ⋆ $S_\star = \{x \in S : m_1(x) = m_2(x) = 1\}$
- **Configuration space** $\Omega_N = \{\boldsymbol{\eta} = (\eta_1, \eta_2) \in \mathbb{N}_0^S \times \mathbb{N}_0^S : |\eta_1| = A,\ |\eta_2| = B\}$
  - ⋆ $\boldsymbol{\eta} = (\boldsymbol{\eta}(x))_{x \in S}, \boldsymbol{\eta}(x) = (\eta_1(x), \eta_2(x))$
  - ⋆ $\rho_N := A/N \to \rho \in (0, 1)$
- **Dynamics** Species i particle jumps $x \to y$ at rate $\boldsymbol{g}_i(\boldsymbol{\eta}(x)) r_i(x, y)$

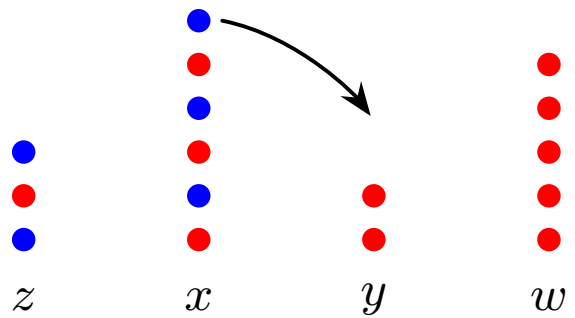
$$\boldsymbol{g}_i(k_1, k_2) = \begin{cases} \frac{k_i}{k_1 + k_2} g(k_1 + k_2) & \text{if} \quad k_1 + k_2 \geq 1 \\ 0 & \text{if} \quad k_1 + k_2 = 0 \end{cases}$$



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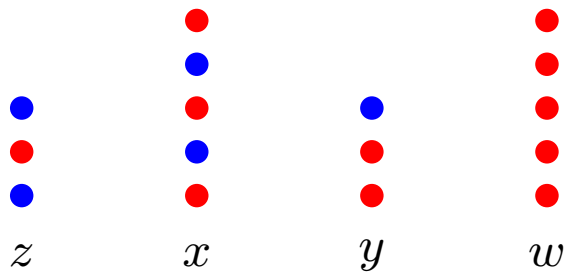
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Condensation

- **Stationary distribution**

$$\nu_N \propto \prod_{x\in S} \binom{|\boldsymbol{\eta}(x)|}{\eta_1(x)} \frac{m_1(x)^{\eta_1(x)} m_2(x)^{\eta_2(x)}}{\mathfrak{a}(|\boldsymbol{\eta}(x)|)}, \quad |\boldsymbol{\eta}(x)| = \eta_1(x) + \eta_2(x).$$

- **Metastable well** For each $x \in S$ and $1 \ll \ell_N \ll N$,

$$\mathcal{E}_N^x := \{\boldsymbol{\eta} : |\boldsymbol{\eta}(x)| \geq N - \ell_N\}, \quad \mathcal{E}_N := \bigcup_{x\in S_\star} \mathcal{E}_N^x$$

- **Negligible set** $\Delta_N := \Omega_N \setminus \mathcal{E}_N$

Theorem (Kim–L. arXiv : 2609.03821)

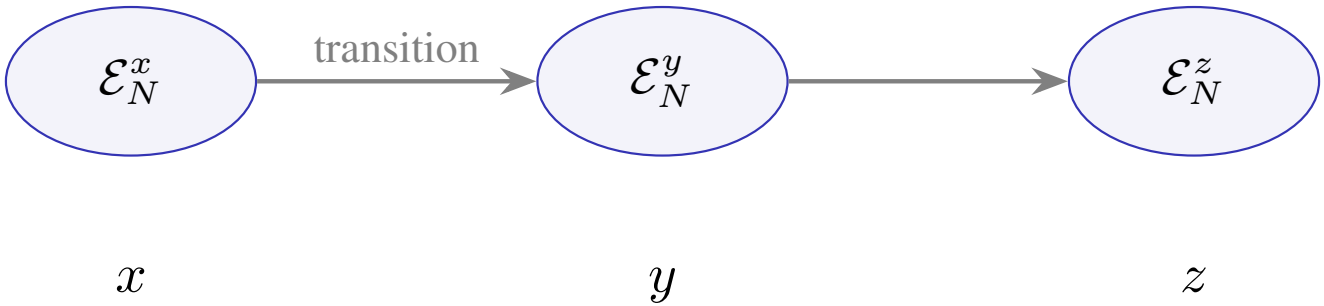
For $\alpha \in (1, \infty)$, we have

$$\lim_{N\to\infty} \nu_N(\mathcal{E}_N^x) = \frac{1}{|S_\star|} \ (x \in S_\star) \quad \text{and} \quad \lim_{N\to\infty} \nu_N(\Delta_N) = 0. \tag{1}$$

Markov chain reduction

- **Trace process $\boldsymbol{\eta}_N^{\mathrm{tr}}$** Obtained by eliminating time spent in Δ_N.
- **Order process X_N**

$$\Psi_N = \sum_{x \in S_\star} x \mathbb{1}_{\mathcal{E}_N^x}, \quad X_N(t) = \Psi_N(\boldsymbol{\eta}_N^{\mathrm{tr}}(t)).$$



Main theorem

- **Metastable time scale** $\theta_N = N^{\alpha+1}$
- **Limiting Markov chain** $\mathbb{X}$ **on** $S_\star$

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha}\left(\frac{\rho}{\mathrm{cap}_1(x,y)} + \frac{1-\rho}{\mathrm{cap}_2(x,y)}\right)^{-1}, \quad x,y \in S_\star.$$

Theorem (Kim–L. arXiv : 2609.03821)

For sufficiently small $\gamma > 0$, *let* $\ell_N = \lfloor \gamma \log N \rfloor$. *Then for* $\alpha > 1$ *and for all* $x \in S_\star$,

1. ***(M) Local metastability*** *For* $b \in (0,1)$

$$\lim_{N\to\infty} \inf_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{P}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x} < H_{\Omega_N \setminus \tilde{\mathcal{E}}_N^x} \right] = 1, \quad \tilde{\mathcal{E}}_N^x = \{\boldsymbol{\eta} \in \Omega_N : |\boldsymbol{\eta}(x)| \geq N - \lfloor N^b \rfloor\}$$

2. ***(C) Convergence*** $\{X_N(\theta_N t)\}_{t\geq 0} \xrightarrow[N\to\infty]{} \{\mathbb{X}(t)\}_{t\geq 0}$.

3. ***(D) Negligibility***

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^T \mathbb{1}\{\boldsymbol{\eta}_N(\theta_N t) \in \Delta_N\} dt \right] = 0, \quad \text{for all} \quad T > 0.$$

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Regime	**Time scale**	**Reversible**	**Non-reversible**
$\alpha > 1$	$\theta_N = N^{\alpha+1}$	Beltran–Landim *PTRF* '12	Seo *CMP* '19
$\alpha = 1$	$\theta_N = N^2 \log N$	Landim–Marcondes–Seo *JEMS* '25	Open

Resolvent Approach

- **Resolvent Approach**: $\forall\, \mathbf{g} : S_\star \to \mathbb{R}$, $\lambda > 0$,

$$(\lambda - \theta_N \mathcal{L}_N) F_N = \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \quad (\lambda - \mathfrak{L}_{\mathbb{X}}) \mathfrak{f} = \mathbf{g},$$

$$\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} : F_N \simeq \mathfrak{f}(x) \quad \text{on each } \mathcal{E}_N^x \ (x \in S_\star) \implies \textbf{(C)+(D)}$$

  Landim–Marcondes–Seo, *J. Eur. Math. Soc.* '25

- $\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} \iff \mathfrak{R}^{(1)} + \mathfrak{R}^{(2)}$:

  ⋆ $\mathfrak{R}^{(1)}$: F_N is asymptotically constant within each well $\mathcal{E}_N^x$:

$$\lim_{N \to \infty} \sup_{\boldsymbol{\eta}, \boldsymbol{\xi} \in \mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi})| = 0, \quad \forall\, x \in S_\star.$$

  ⋆ $\mathfrak{R}^{(2)}$: Let $f_N(x) = \mathbb{E}_{\nu_N}[F_N \mid \mathcal{E}_N^x]$, then $f_N(x) \to \mathfrak{f}(x)$.

- $\mathfrak{R}^{(1)}$: Suffices to show that

$$\lim_{N \to \infty} \sup_{\boldsymbol{\eta} \in \mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi}_N^x)| = 0.$$

$$(\lambda - \theta_N \mathcal{L}_N) F_N = \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}$$

Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^*$:

$$\begin{cases} \mathcal{L}_N^* \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \Omega_N \setminus \mathcal{E}_N \\ \mathfrak{H}_x^* \equiv 1 & \text{on} \quad \mathcal{E}_N^x \\ \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \mathcal{E}_N \setminus \mathcal{E}_N^x \end{cases}, \quad \mathfrak{H}_x^*(\boldsymbol{\eta}) := \mathbb{P}_{\boldsymbol{\eta}}^{N,*}[H_{\mathcal{E}_N^x} < H_{\mathcal{E}_N \setminus \mathcal{E}_N^x}].$$

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$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \left\langle \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \mathfrak{H}_x^* \right\rangle_{\nu_N}$$

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$$\Downarrow$$

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathfrak{H}_x^* \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

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$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) - \nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

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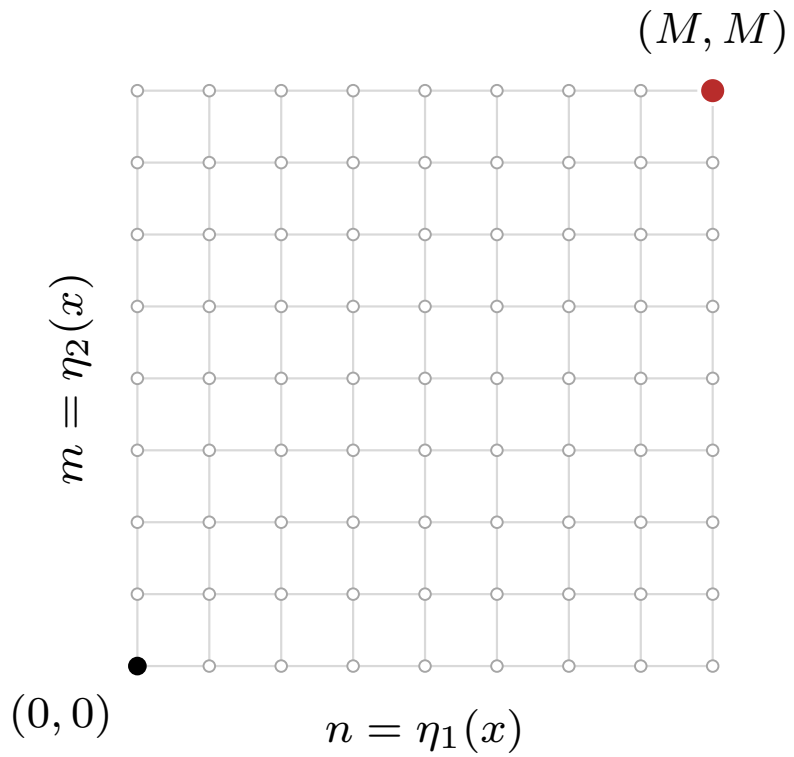
$$(\lambda - \mathfrak{L}_{\mathbb{X}}) f_N(x) = \mathbf{g}(x) + o(1)$$

$$\mathbb{W}_x|_{\mathcal{E}_N^x} \equiv 1, \quad \mathbb{W}_x|_{\mathcal{E}_N \setminus \mathcal{E}_N^x} \equiv 0, \quad \theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N} \simeq \theta_N \langle -\mathcal{L}_N F_N, \mathbb{W}_x \rangle_{\nu_N}$$

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



Two condensates

Both species are at x:

$$(n, m) = (M, M).$$

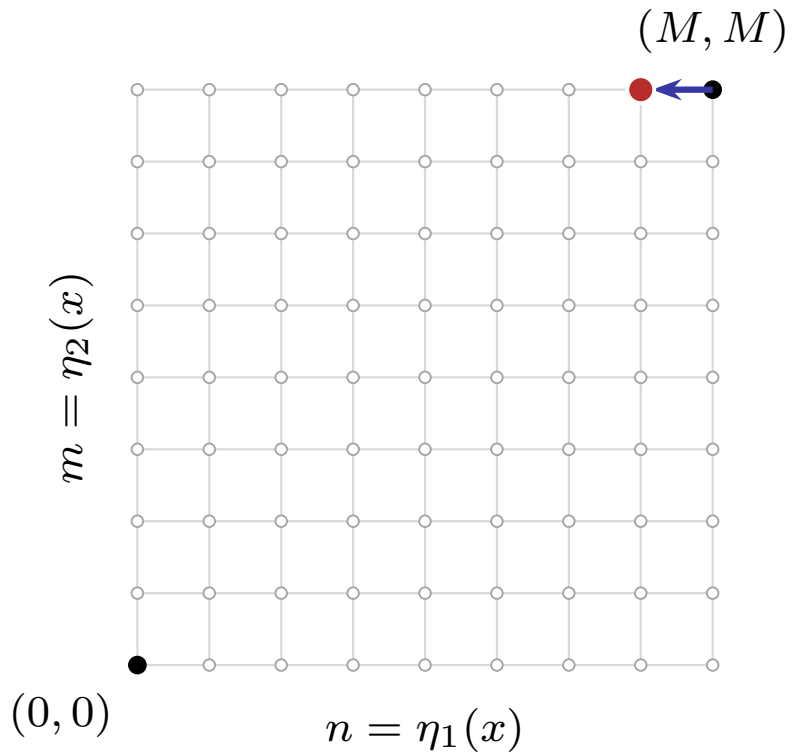
The other corner $(0, 0)$ represents condensation at y.

We seek a test function equal to 1 and 0 at these corners.

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A single-particle jump

One species-1 particle moves from x to y:

$$(M, M) \longrightarrow (M - 1, M).$$

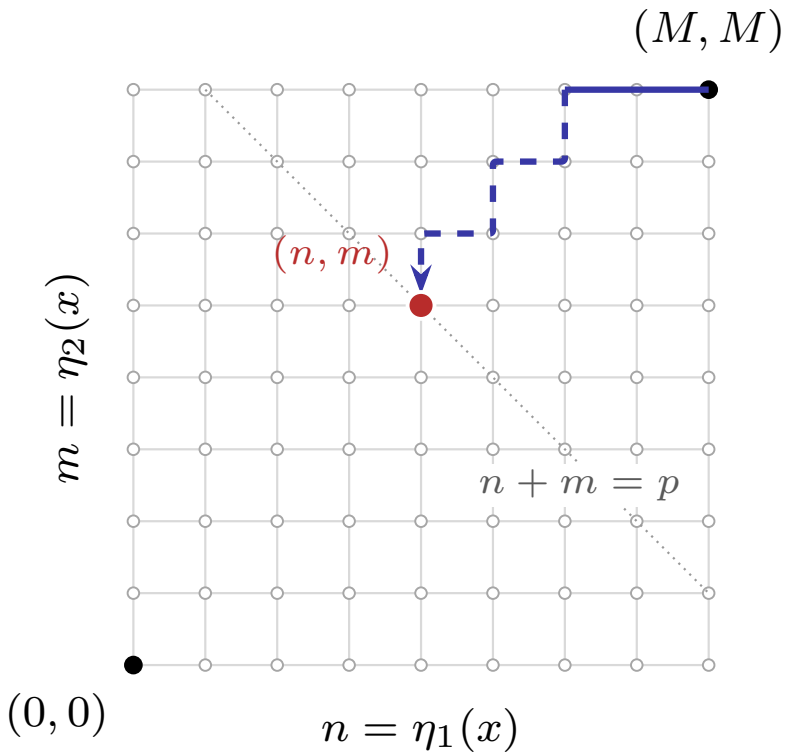
Its jump rate is

$$\tfrac{1}{2} g(2M)\, r_1(x, y).$$

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An intermediate configuration

Stop at a state with

$$p = n + m$$

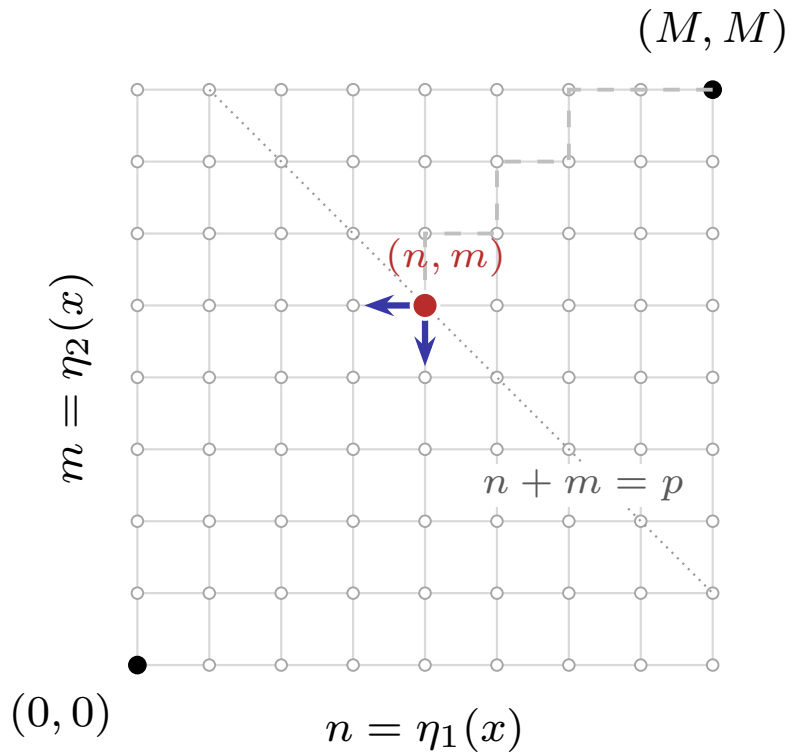
$$\eta(x) = (n, m), \quad \eta(y) = (M-n, M-m).$$

The dashed segment is an illustrative sequence of $x \to y$ jumps.

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Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



Two ways to transfer one particle

For the $x \to y$ jumps at $p = n + m$,

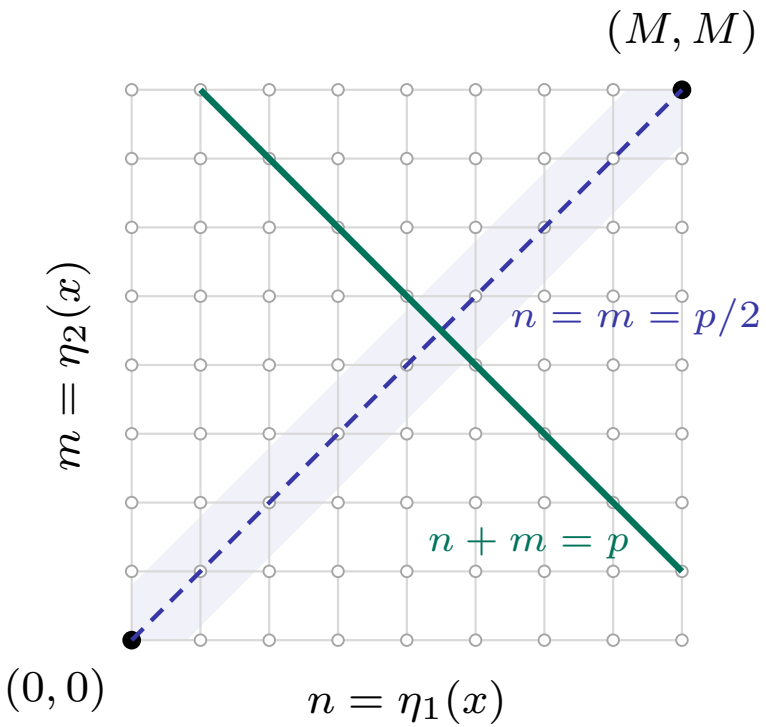
$$q_{\leftarrow} = \frac{n}{p} \cdot \frac{p^{\alpha}}{(p-1)^{\alpha}} r_1(x, y),$$

$$q_{\downarrow} = \frac{m}{p} \cdot \frac{p^{\alpha}}{(p-1)^{\alpha}} r_2(x, y).$$

Both lower p by one, but change $n - m$ by -1 and $+1$, respectively.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



A total-mass profile

The one-dimensional energy estimate suggests

$$H_p = \frac{\sum_{k=0}^{p-1} \mathfrak{a}(k)\mathfrak{a}(N-k-1)}{\sum_{k=0}^{N-1} \mathfrak{a}(k)\mathfrak{a}(N-k-1)}.$$

Thus $H_0 = 0$ and $H_N = 1$.

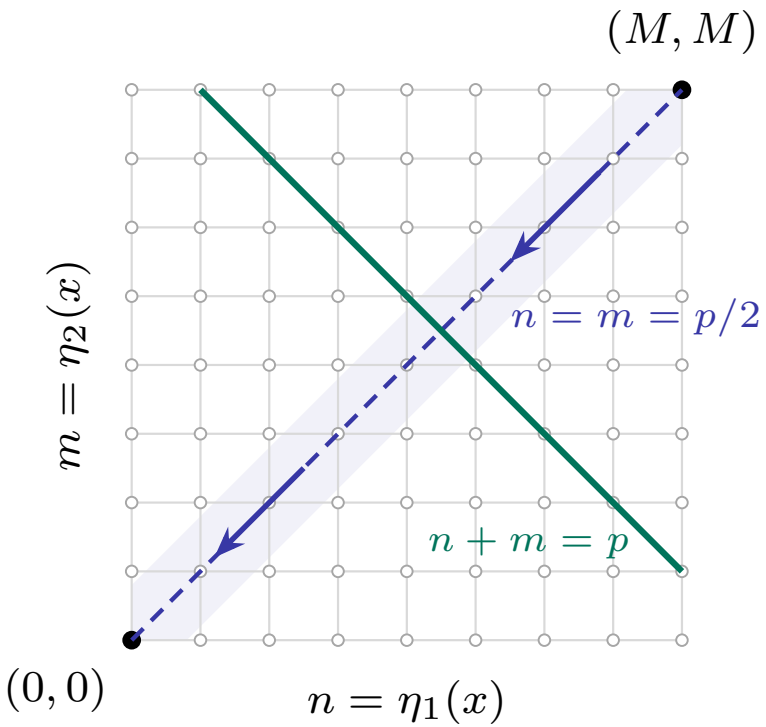
If $R_1 \neq R_2$, optimal increments in the two species directions differ:

$$\nabla_1 G : \nabla_2 G \approx R_1^{-1} : R_2^{-1}.$$

A function of p alone has equal increments in both directions.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



Conditional composition

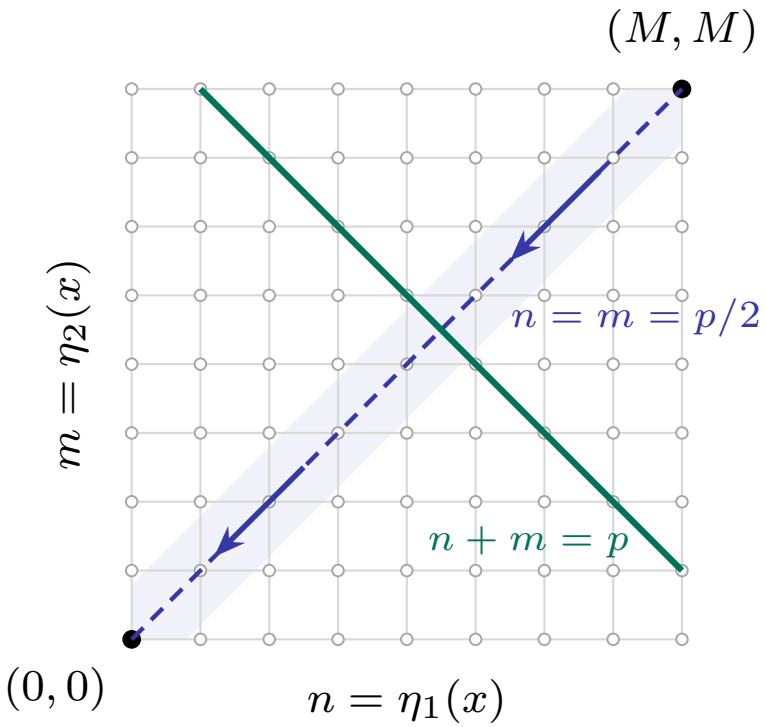
Under the stationary law, given $n + m = p$,

$$\nu_N(n, p - n \mid p) = \frac{\binom{p}{n}\binom{N-p}{M-n}}{\binom{N}{M}}.$$

Deviation

$$n - \frac{p}{2} = \frac{n - m}{2}, \quad \mathbb{E}[n \mid p] = \frac{p}{2}.$$

Reduced model: why a composition corrector?



Correcting for unequal rates

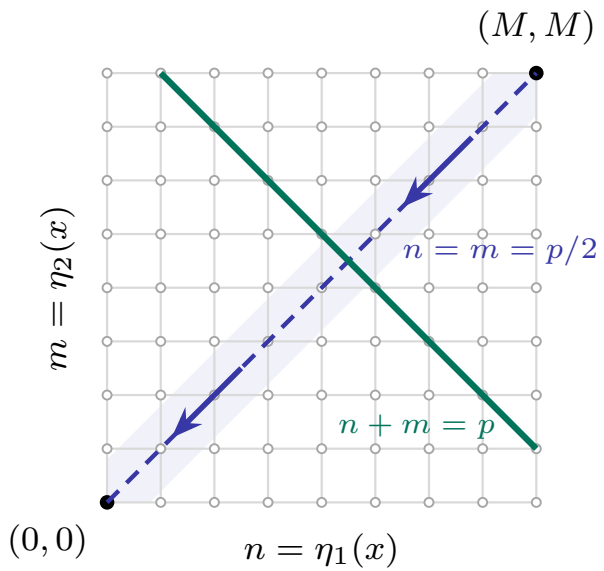
$$\boxed{G(n,m) = H_p + \beta_p \left(n - \frac{p}{2} \right)}$$

$$\beta_p = \frac{H_p - H_{p-1}}{\vartheta} \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \beta_0 = 0,$$

$$\vartheta = \frac{1}{2R_1} + \frac{1}{2R_2}.$$

The correction has zero conditional mean and adjusts the two directional increments.

Reduced model: why a composition corrector?



Correcting for unequal rates

$$G(n, m) = H_p + \beta_p \left(n - \frac{p}{2} \right)$$

We extend this corrected profile to the full configuration space.

Global test function on Ω_N

For each $x \in S_\star$,

$$\mathbb{W}_x(\boldsymbol{\eta}) = \sum_{y \in S_\star \setminus \{x\}} \Theta_\epsilon^{x,y}(\boldsymbol{\eta}/N)\, \mathcal{W}_{x,y}(\boldsymbol{\eta}),$$

$$\mathbb{W}_x\big|_{\mathcal{E}_N^x} \equiv 1, \qquad \mathbb{W}_x\big|_{\mathcal{E}_N \setminus \mathcal{E}_N^x} \equiv 0, \qquad \mathbb{W}_x\big|_{\text{between } \mathcal{E}_N^x, \mathcal{E}_N^y} = \mathcal{W}_{x,y}.$$

Completing $\mathfrak{R}^{(2)}$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \left\langle \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \mathbb{W}_x \right\rangle_{\nu_N}$$

Completing $\mathfrak{R}^{(2)}$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \left\langle \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \mathbb{W}_x \right\rangle_{\nu_N}$$

$$\Downarrow$$

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

and

$$\begin{aligned} \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} &= \nu_N(\mathcal{E}_N^x) \sum_{y \neq x} \mathfrak{r}(x,y)[f_N(x) - f_N(y)] + o(1) \\ &= -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) + o(1). \end{aligned}$$

The $\mathfrak{R}^{(2)}$ condition

Let $(\lambda - \mathfrak{L}_{\mathbb{X}})\mathfrak{f} = \mathbf{g}$, then we obtain

$$\|f_N - \mathfrak{f}\|_\infty \xrightarrow[N \to \infty,\ \epsilon \downarrow 0]{} 0.$$

Further topics

Q. Generalize for super-critical k-species case?

With time scale $\theta_N = N^{\alpha+1}$,

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha} \left(\sum_{j=1}^{k} \frac{\rho_j}{\mathrm{cap}_j(x,y)} \right)^{-1}$$

Q. Can we obtain analogous results at $\alpha = 1$?

The difficult part is the local metastability for larger wells.

Thank you!